IB PHYSICS STUDENT WORKSHEET
THE OVERLOADED CARGO SHIP

Buoyancy • Density • Archimedes' Principle • Equilibrium

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Learning goal: I can use density, displacement, and Archimedes' Principle to explain why a large steel ship can float and decide what happens when its mass becomes too large.

IB Physics skills:

1. Identifying a system and its boundary
2. Selecting equations, using units, interpreting results & explaining with physics

1. QUICK PHYSICS TOOLKIT

Density	**ρ = m / V**
Buoyant force	$F_\beta = \rho_{fluid.} V_{displaced.}\, g$ (ρ.g.V)
Weight	W = mg
Floating equilibrium	$F_\beta = W$

Remember: A steel block sinks because steel is denser than water. A ship can float because we treat the whole ship as a system: its steel structure encloses a large volume, so the average density of the whole system can be less thanc the density of water.

2. INQUIRY HOOK (THINK BEFORE CALCULATING)

A solid block of steel has a density of about 7850 kg m^{-3}. Seawater has a density of about 1025 kg m^{-3}. Yet a huge ship made mostly from steel can float.

Question: How is this possible? Explain using the ideas of average density and the volume enclosed by the ship.

A ship is not fully solid still, it has hulls that encloses a large volume of air/empty space. Referring to prior formula, the average density of the whole ship is:

$$\rho_{avg} = \frac{m_{total}}{V_{total}}$$

The air adds a lot of volume but adds small on mass, according formula ρ_{avg} will decrease to the point that it is less than seawater density, thus the ship floats when the buoyant force equal to its weight

3. BUILD THE MODEL

For this worksheet, treat the ship as one system. Its rigid hull has a fixed external volume. Cargo and water taken into the ship increase total mass, but the model assumes the hull volume stays constant.

*Hull: The **hull** is the main body of a boat or yacht that touches the water. In simple words, it is the foundation of the vessel, giving it shapes strength, and balance. Without the hull, a boat cannot float or move safely.

a) The ship's total mass is made up of steel, loads, and air but air can be negligible

b) In our simplified model, the ship's volume is fixed

c) If mass increases while volume stays constant, average density will increase

Key equation: $\rho_{avg} = m_{total} / V$

$$\rho_{\text{avg}} = \frac{m_{\text{total}}}{V_{\text{total}}} = \frac{m_{\text{steel}} + m_{\text{cargo}} + m_{\text{air}}}{V_{\text{hull}}}$$

4. CASE STUDY (MV TITAN)

MV Titan is a large cargo vessel travelling through seawater. Use the information below to investigate whether the ship can remain afloat.

Initial structural + equipment mass	**7.68×10^7 kg**
Fixed hull volume	214 400 m^3
Seawater density	1025 kg m^{-3}
Overloaded total mass	2.50×10^8 kg
Model assumption	Hull volume remains constant

5. PART A (SYSTEM BOUNDARY)

When cargo is added, does the geometric volume of the rigid hull change in this model? Circle one: **NO**

Explain your choice. Use the words system, mass, and volume.

The system is the ship plus cargo plus trapped air. Adding cargo increases the system's mass, but the rigid hull's external volume is assumed constant in this model.

6. PART B (INITIAL AVERAGE DENSITY)

Use:

$$\rho_{\text{avg}} = \frac{m_{\text{total}}}{V_{\text{total}}} = \frac{m_{\text{steel}} + m_{\text{cargo}} + m_{\text{air}}}{V_{\text{hull}}}$$

Step 1. Write the values with units:

$m = 7.68 \times 10^{7} kg$

$V = 214400\ m^{3}$

$\rho_{seawater} = 1025\ kg\ m^{-3}$

Step 2. Substitute into the equation:

$\rho_{avg} = \frac{m}{V}$

$\rho_{avg} = \frac{7.68 \times 10^{7}}{214400}$

Step 3. Calculate and give your answer to 3 significant figures:

$\rho_{avg} = 358\ kg\ m^{-3}$

Compare your result with seawater density. Is the initial ship's average density greater than or less than seawater? What does this suggest about its ability to float?

$358\ kg\ m^{-3} < 1025\ kg\ m^{-3}$

This suggests it can float, with spare buoyancy

7. PART C (THE OVERLOADED SHIP)

During severe weather, MV Titan takes on water and also carries additional heavy cargo. Its total mass becomes 2.50×10^{8} kg. The model still uses the same hull volume: 214 400 m³.

Calculate the new average density. Show your working clearly.

Equation:

$\rho_{avg} = \frac{m}{V}$

Substitution:

$m = 2.50 \times 10^{8} kg$

$$V = 214400\ m^3$$

$$\rho_{avg} = \frac{2.50 \times 10^8}{214400}$$

Final answer:

$$\rho_{avg} = 1166\ kg\ m^{-3}$$

$$\rho_{avg} \approx 1.17 \times 10^3 kg\ m^{-3}$$

Compare your new average density with 1025 kg m^{-3} seawater.

$$1.17 \times 10^3 kg\ m^{-3} > 1025\ kg\ m^{-3}$$

8. PART D (WHAT HAPPENS TO THE FORCES?)

At equilibrium, the buoyant force must equal the ship's weight: $F_\beta = W$

• The ship's weight is W = mg, so when mass increases, the weight increases

• The maximum buoyant force available depends on the maximum displaced volume, seawater density, and gravity

• Because the overloaded average density is greater than seawater density, $F_\beta < W$

• Therefore, hydrostatic equilibrium **cannot** be maintained, so the ship sinks

Final prediction: **SINK**

$$F_{\beta\,max} = \rho_{seawater} mg$$

$$F_{\beta\,max} = 1025(214400)(9.8)$$

$$F_{\beta\,max} \approx 2.16 \times 10^9 N$$

$$W = mg$$

$$W = (2.50 \times 10^8)(9.8)$$

$$W \approx 2.45 \times 10^9 N$$

$W > F_{\beta\, max}$, the ship sinks

9. IB (STYLE CONCEPT CHECK)

A. Why does the steel itself not need to be less dense than seawater for the ship to float?

The steel is only a part of the ship and the ship itself is not a solid block of steel. The steel is shaped into a hull that traps a large volume of air. The more accurate density to be considered is the average density of the whole ship, not only the steel. Thus, according to buoyant force if

$\rho_{avg} < \rho_{seawater}$

The ship will float

B. Does the trapped air provide a separate upward force that 'pushes' the cargo upward? Explain.

No, the force is provided by buoyant force. But also because of the air, the average density of the ship will decrease while the increase of mass is negligible and helps to satisfy

$\rho_{avg} < \rho_{seawater}$

So that the ship can float

C. What changes when mass is added but the ship's volume stays constant?

The weight increases, resulting in higher average density, the ship must displace mroe water to stay afloat

D. Why can a ship stop floating even though its hull volume has not changed?

Because the mass added is too much that exceeds the maximum possible buoyant force to keep the ship afloat

10. EXTENSION (HOW MUCH MASS CAN THE SHIP SUPPORT?)

For the simplified model, the largest possible displaced volume is 214 400 m³. At the limit of floating:

$\boldsymbol{\rho_{seawater} \cdot V = m_{limit}}$

Calculate the maximum total mass that can be supported by the model:

$\rho_{seawater} V = m_{limit}$

$m_{limit} = 1025 \times 214400$

$m_{limit} = 219760000\ kg$

$m_{limit} \approx 2.20 \times 10^{8} kg$

How does this compare with the overloaded mass of 2.50×10^{8} kg?

$2.50 \times 10^{8} kg > 2.20 \times 10^{8} kg$

The overloaded mass is too large, exceeding the maximum supported mass so the ship will sink/stop floating

11. EXIT REFLECTION

The most important idea I learned today is…

The buoyancy force, the very concept for every floating ships

One physics term I can now use correctly is…

Average density/buoyant force/Archimedes' Principle

One step in my calculation/reasoning that I need to improve is…

Keeping the system boundary clear and using correct units consistently

IB habit of mind: Don't just answer, use physics to justify it.